

# Test and Configure ROS2

① Before starting this course, please refer to "2.Linux Basic Lesson\Lesson 3 Linux Installation and Source Replacement" to install the virtual machine client and import the ROS2 Ubuntu image.

② The image file we provide is in the "6.Appendix/2.Virtual Machine Image\Hiwonder\_ROS\_accessory\_ros2" folder, and the ROS2 environment is already configured. You can refer to this section for a brief overview of the ROS2 system installation and environment setup.

## 1. Version Options

Different versions of ROS2 are installed on different versions of Ubuntu. In this section, we will use the virtual machine with Ubuntu 20.04, which corresponds to ROS2 Foxy.

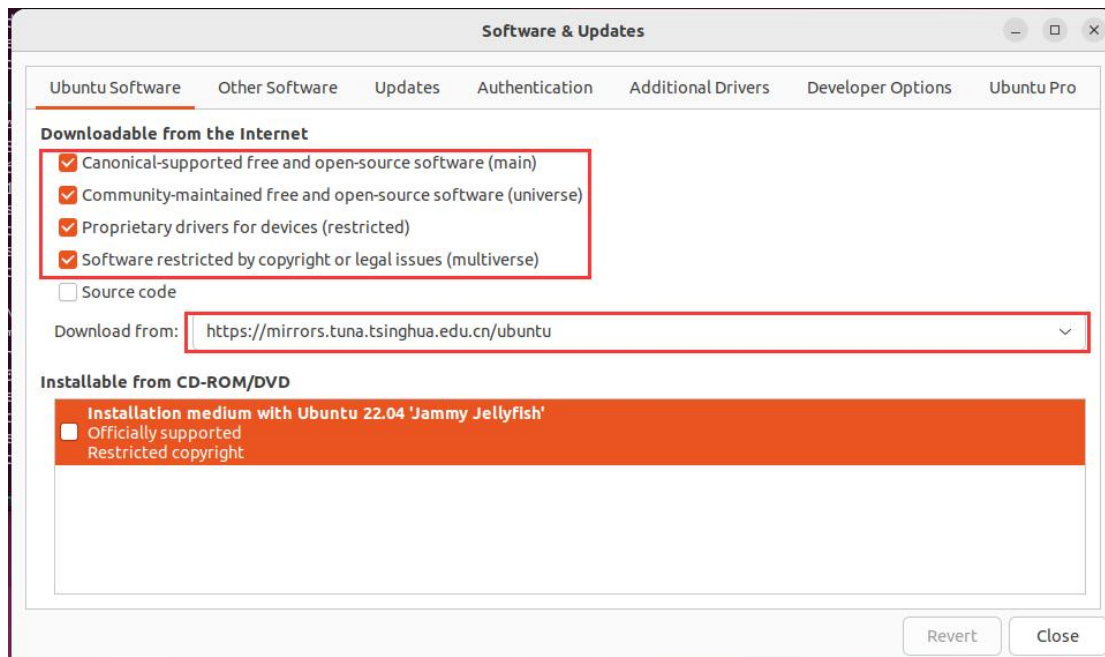
| ROS2版本              | 发布时间    | 维护截止时间  | Ubuntu版本                                                   |
|---------------------|---------|---------|------------------------------------------------------------|
| Ardent Apalone      | 2017.12 | 2018.12 | Ubuntu 16.04 (Xenial Xerus)                                |
| Bouncy Bolson       | 2018.7  | 2019.7  | Ubuntu 16.04 (Xenial Xerus) 、 Ubuntu 18.04 (Bionic Beaver) |
| Crystal Clemmys     | 2018.12 | 2019.12 | Ubuntu 18.04 (Bionic Beaver)                               |
| Dashing Diademata   | 2019.5  | 2021.5  | Ubuntu 18.04 (Bionic Beaver)                               |
| Eloquent Elusor     | 2019.11 | 2020.11 | Ubuntu 18.04 (Bionic Beaver)                               |
| Foxy Fitzroy        | 2020.6  | 2023.5  | Ubuntu 20.04 (Focal Fossa)                                 |
| Galactic Geochelone | 2021.5  | 2022.11 | Ubuntu 20.04 (Focal Fossa)                                 |
| Humble Hawksbill    | 2022.5  | 2027.5  | Ubuntu 22.04 (Jammy Jellyfish)                             |

## 2. Check Ubuntu's Software and Update Sources

1) Locate 'System&Update'.



- 2) Make sure to check the option for internet downloads and change the download source to the local region's source.



### 3. Set Encoding Format

- 1) Enable the virtual machine, and press "Ctrl+Alt+T" to open the terminal.
- 2) Input

```
sudo locale-gen en_US en_US.UTF-8
```

Set the Ubuntu locale to en\_US.UTF-8. After entering the command, press Enter, then enter the virtual machine's password, "**hiwonder**" (users should enter the password they have set for their own virtual machine; in our case, the password is "**hiwonder**").

```
ubuntu@ubuntu-virtual-machine:~$ sudo locale-gen en_US en_US.UTF-8
[sudo] password for ubuntu:
Generating locales (this might take a while)...
  en_US.ISO-8859-1... done
  en_US.UTF-8... done
Generation complete.
ubuntu@ubuntu-virtual-machine:~$
```

- 3) Run the command below and press Enter key to update the language environment.

```
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
```

```
ubuntu@ubuntu-virtual-machine:~$ sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
ubuntu@ubuntu-virtual-machine:~$
```

- 4) Enter the command, press Enter, and export the environment variables to take effect immediately.

```
export LANG=en_US.UTF-8
```

```
ubuntu@ubuntu-virtual-machine:~$ export LANG=en_US.UTF-8
ubuntu@ubuntu-virtual-machine:~$
```

## 4. Set the Download Source for ROS2

### 4.1 Enable the Ubuntu Universe Repository

- 1) Enter the command to check if the Universe repository is enabled:

```
apt-cache policy | grep universe
```

```
ubuntu@ubuntu-virtual-machine:~$ apt-cache policy | grep universe
```

- 2) If the output is as follows, it indicates that it has been enabled, and you can proceed to configure it further in section 1.4.1.

```
500 http://mirrors.aliyun.com/ubuntu focal-security/universe i386 Packages
      release v=20.04,o=Ubuntu,a=focal-security,n=focal,l=Ubuntu,c=universe,b=i3
86
500 http://mirrors.aliyun.com/ubuntu focal-security/universe amd64 Packages
      release v=20.04,o=Ubuntu,a=focal-security,n=focal,l=Ubuntu,c=universe,b=am
d64
100 http://mirrors.aliyun.com/ubuntu focal-backports/universe i386 Packages
      release v=20.04,o=Ubuntu,a=focal-backports,n=focal,l=Ubuntu,c=universe,b=i
386
100 http://mirrors.aliyun.com/ubuntu focal-backports/universe amd64 Packages
      release v=20.04,o=Ubuntu,a=focal-backports,n=focal,l=Ubuntu,c=universe,b=a
md64
500 http://mirrors.aliyun.com/ubuntu focal-updates/universe i386 Packages
      release v=20.04,o=Ubuntu,a=focal-updates,n=focal,l=Ubuntu,c=universe,b=i38
6
```

- 3) If you see the following prompt, it means it has not been enabled.

```
ubuntu@ubuntu-virtual-machine:~$ apt-cache policy | grep universe
E: Cache is out of sync, can't x-ref a package file
```

- 4) Enter the command to install the software package  
"software-properties-common":

## sudo apt install software-properties-common

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt install software-properties-common
Reading package lists... Done
Building dependency tree
Reading state information... Done
software-properties-common is already the newest version (0.99.9.8).
software-properties-common set to manually installed.
The following packages were automatically installed and are no longer required:
```

- 5) Enter the command to add the PPA to the Universe repository and automatically import the public key:

## sudo add-apt-repository universe

```
ubuntu@ubuntu-virtual-machine:~$ sudo add-apt-repository universe
'universe' distribution component is already enabled for all sources.
```

- 6) You can enter the command again to check if it has been enabled:

## apt-cache policy | grep universe

```
ubuntu@ubuntu-virtual-machine:~$ apt-cache policy | grep universe
500 http://mirrors.aliyun.com/ubuntu focal-security/universe i386 Packages
  release v=20.04,o=Ubuntu,a=focal-security,n=focal,l=Ubuntu,c=universe,b=i3
86
500 http://mirrors.aliyun.com/ubuntu focal-security/universe amd64 Packages
  release v=20.04,o=Ubuntu,a=focal-security,n=focal,l=Ubuntu,c=universe,b=am
d64
100 http://mirrors.aliyun.com/ubuntu focal-backports/universe i386 Packages
  release v=20.04,o=Ubuntu,a=focal-backports,n=focal,l=Ubuntu,c=universe,b=i
386
100 http://mirrors.aliyun.com/ubuntu focal-backports/universe amd64 Packages
  release v=20.04,o=Ubuntu,a=focal-backports,n=focal,l=Ubuntu,c=universe,b=a
md64
500 http://mirrors.aliyun.com/ubuntu focal-updates/universe i386 Packages
  release v=20.04,o=Ubuntu,a=focal-updates,n=focal,l=Ubuntu,c=universe,b=i38
6
```

## 4.2 Update the software packages and install the curl and gnupg tools

- 1) Enter the command to update the software installation packages:

## sudo apt update

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt update
Hit:1 http://mirrors.aliyun.com/ubuntu focal InRelease
Get:2 http://mirrors.aliyun.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://mirrors.aliyun.com/ubuntu focal-backports InRelease [108 kB]
Get:4 http://mirrors.aliyun.com/ubuntu focal-security InRelease [114 kB]
Get:5 http://mirrors.aliyun.com/ubuntu focal/main DEP-11 64x64@2 Icons [15.8 kB]
1
```



- 2) After the update is complete, it will appear as shown in the following image.

```
Get:29 http://mirrors.aliyun.com/ubuntu focal-security/multiverse amd64 DEP-11
Metadata [2,464 B]
Get:30 http://mirrors.aliyun.com/ubuntu focal-security/multiverse DEP-11 64x64@
2 Icons [29 B]
Fetched 5,900 kB in 46s (127 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
ubuntu@ubuntu-virtual-machine:~$
```

- 3) Enter the command to install the curl and gnupg tools:

```
sudo apt install curl gnupg2 lsb-release
```

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt install curl gnupg2 lsb-release
Reading package lists... Done
Building dependency tree
Reading state information... Done
lsb-release is already the newest version (11.1.0ubuntu2).
lsb-release set to manually installed.
```

- 4) During the installation process, if you see the prompt as shown in the following image, simply enter "y" and press Enter.

```
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  curl gnupg2
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 167 kB of archives.
After this operation, 464 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

- 5) After the installation is complete, it will appear as shown in the following image:

```
Get:2 http://mirrors.aliyun.com/ubuntu focal-updates/universe amd64 gnupg2 all
2.2.19-3ubuntu2.2 [5,316 B]
Fetched 167 kB in 2s (77.9 kB/s)
Selecting previously unselected package curl.
(Reading database ... 225190 files and directories currently installed.)
Preparing to unpack .../curl_7.68.0-1ubuntu2.14_amd64.deb ...
Unpacking curl (7.68.0-1ubuntu2.14) ...
Selecting previously unselected package gnupg2.
Preparing to unpack .../gnupg2_2.2.19-3ubuntu2.2_all.deb ...
Unpacking gnupg2 (2.2.19-3ubuntu2.2) ...
Setting up gnupg2 (2.2.19-3ubuntu2.2) ...
Setting up curl (7.68.0-1ubuntu2.14) ...
Processing triggers for man-db (2.9.1-1) ...
ubuntu@ubuntu-virtual-machine:~$
```

## 4.3 Set the public key

Enter the following command:

```
curl -s https://raw.githubusercontent.com/ros/rosdistro/master/ros.asc |  
sudo apt-key add -
```

```
ubuntu@ubuntu-virtual-machine:~$ curl -s https://raw.githubusercontent.com/ros/rosdistro/master/ros.asc | sudo apt-key add -  
OK  
ubuntu@ubuntu-virtual-machine:~$
```

If the following error appears, re-enter the command:

```
curl http://repo.ros2.org/repos.key | sudo apt-key add -
```

```
hiwonder@ubuntu:~$ curl -s https://raw.githubusercontent.com/ros/rosdistro/master/ros.asc | sudo apt-key add -  
[sudo] password for hiwonder:  
gpg: no valid OpenPGP data found.  
hiwonder@ubuntu:~$
```

## 4.4 Set Download Source

Enter the following command:

```
sudo sh -c 'echo "deb [arch=$(dpkg --print-architecture)]  
http://packages.ros.org/ros2/ubuntu $(lsb_release -cs) main" >  
/etc/apt/sources.list.d/ros2-latest.list'
```

```
ubuntu@ubuntu-virtual-machine:~$ sudo sh -c 'echo "deb [arch=$(dpkg --print-architecture)] http://packages.ros.org/ros2/ubuntu $(lsb_release -cs) main" > /etc/apt/sources.list.d/ros2-latest.list'  
ubuntu@ubuntu-virtual-machine:~$
```

## 5. Install ROS2

1) Enter the below command to update the software feature pack.

```
sudo apt update
```

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt update
Hit:1 http://mirrors.aliyun.com/ubuntu focal InRelease
Get:2 http://mirrors.aliyun.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://mirrors.aliyun.com/ubuntu focal-backports InRelease [108 kB]
Get:4 http://packages.ros.org/ros2/ubuntu focal InRelease [4,685 B]
Get:5 http://mirrors.aliyun.com/ubuntu focal-security InRelease [114 kB]
Get:6 http://packages.ros.org/ros2/ubuntu focal/main amd64 Packages [1,151 kB]
```

2) After the update is complete, it will appear as shown below.

```
Get:16 http://mirrors.aliyun.com/ubuntu focal-security/multiverse amd64 DEP-11
Metadata [2,468 B]
Fetched 5,288 kB in 12s (427 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
ubuntu@ubuntu-virtual-machine:~$
```

3) Run the below command:

**sudo apt install ros-foxy-desktop**

Press Enter. When prompted whether to install, type "y" and press Enter again. Wait for the download and installation to complete. Depending on your network conditions, this process may take about 10 minutes, so please be patient.

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt install ros-foxy-desktop
Reading package lists... Done
Building dependency tree
Reading state information... Done
x11proto-input-dev x11proto-randr-dev x11proto-scrnsaver-dev
x11proto-xext-dev x11proto-xf86vidmode-dev x11proto-xinerama-dev
xorg-sgml-doctools xtrans-dev zlib1g-dev
0 upgraded, 953 newly installed, 0 to remove and 0 not upgraded.
Need to get 475 MB of archives.
After this operation, 2,668 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

4) After the installation is complete, it will appear as shown below.

```
Setting up ros-foxy-geometry2 (0.13.13-1focal.20221013.004359) ...
Setting up ros-foxy-rviz-common (8.2.7-1focal.20221013.003214) ...
Setting up ros-foxy-rqt-common-plugins (1.0.0-1focal.20221013.012401) ...
Setting up ros-foxy-ros-base (0.9.2-1focal.20221013.012938) ...
Setting up ros-foxy-rviz-default-plugins (8.2.7-1focal.20221013.012643) ...
Setting up ros-foxy-rviz2 (8.2.7-1focal.20221013.020616) ...
Setting up ros-foxy-desktop (0.9.2-1focal.20221013.021123) ...
Processing triggers for libc-bin (2.31-0ubuntu9.9) ...
ubuntu@ubuntu-virtual-machine:~$
```



## 6. Install Additional Dependency

- 1) Enter the command, press Enter, and input the virtual machine password "hiwonder" to install the autocomplete tool:

**sudo apt install python3-argcomplete**

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt install python3-argcomplete
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
 adium-theme-ubuntu command-not-found-data diffstat enchant example-content
 fwupdate geoip-database gnome-software-common gnome-user-guide
 guile-2.0-libs ifupdown imagemagick-6-common iputils-arping
 libappstream-glib8 libarchive-zip-perl libargon2-0 libart-2.0-2
```

- 2) After the installation is complete, it will appear as shown below.

```
Selecting previously unselected package python3-argcomplete.
(Reading database ... 285089 files and directories currently installed.)
Preparing to unpack ../python3-argcomplete_1.8.1-1.3ubuntu1_all.deb ...
Unpacking python3-argcomplete (1.8.1-1.3ubuntu1) ...
Setting up python3-argcomplete (1.8.1-1.3ubuntu1) ...
Processing triggers for man-db (2.9.1-1) ...
ubuntu@ubuntu-virtual-machine:~$
```

- 3) Input the following command and hit Enter key to install the compilation tool. Then enter 'y' to confirm the installation.

**sudo apt install python3-colcon-common-extensions**

```
ubuntu@ubuntu-virtual-machine:~$ sudo apt install python3-colcon-common-extensions
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

```
0 upgraded, 35 newly installed, 0 to remove and 0 not upgraded.
Need to get 766 kB of archives.
After this operation, 4,769 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

- 4) After the installation is complete, it will appear as shown below.



```
Setting up python3-colcon-recursive-crawl (0.2.1-1) ...
Setting up python3-colcon-cmake (0.2.27-1) ...
Setting up python3-colcon-notification (0.2.14-1) ...
Setting up python3-colcon-cd (0.1.1-1) ...
Setting up python3-colcon-ros (0.3.23-1) ...
Setting up python3-colcon-common-extensions (0.3.0-1) ...
Processing triggers for man-db (2.9.1-1) ...
ubuntu@ubuntu-virtual-machine:~$
```

## 7. Set Environment Variables

Sequentially enter the following commands, pressing Enter after each one, to complete the environment variable setup:

```
echo "source /opt/ros/foxy/setup.bash" >> ~/.bashrc
```

```
source ~/.bashrc
```

```
ubuntu@ubuntu-virtual-machine:~$ echo "source /opt/ros/foxy/setup.bash" >> ~/.bashrc
ubuntu@ubuntu-virtual-machine:~$ source ~/.bashrc
```

## 8. Test ROS2 System

1) Enter the command to run a talker.

```
ros2 run demo_nodes_cpp talker
```

```
ubuntu@ubuntu-virtual-machine:~$ ros2 run demo_nodes_cpp talker
[INFO] [1667186428.542832588] [talker]: Publishing: 'Hello World: 1'
[INFO] [1667186429.542654738] [talker]: Publishing: 'Hello World: 2'
[INFO] [1667186430.542757313] [talker]: Publishing: 'Hello World: 3'
[INFO] [1667186431.542860113] [talker]: Publishing: 'Hello World: 4'
[INFO] [1667186432.542704202] [talker]: Publishing: 'Hello World: 5'
[INFO] [1667186433.543069939] [talker]: Publishing: 'Hello World: 6'
```

2) Press "Ctrl+Alt+T" to open a new terminal, then enter the command to run a listener.

```
ros2 run demo_nodes_py listener
```

```
ubuntu@ubuntu-virtual-machine:~$ ros2 run demo_nodes_py listener
[INFO] [1667186523.213157479] [listener]: I heard: [Hello World: 11]
[INFO] [1667186524.197304126] [listener]: I heard: [Hello World: 12]
[INFO] [1667186525.196535619] [listener]: I heard: [Hello World: 13]
[INFO] [1667186526.196780338] [listener]: I heard: [Hello World: 14]
[INFO] [1667186527.196269022] [listener]: I heard: [Hello World: 15]
[INFO] [1667186528.196268775] [listener]: I heard: [Hello World: 16]
[INFO] [1667186529.196815578] [listener]: I heard: [Hello World: 17]
[INFO] [1667186530.195096077] [listener]: I heard: [Hello World: 18]
```

- 3) Press "Ctrl+C" to stop the execution. If stopping fails, you can repeat pressing "Ctrl+C" multiple times until it stops.